

Introduction to linear regression

The Human Freedom Index is a report that attempts to summarize the idea of “freedom” through a bunch of different variables for many countries around the globe. It serves as a rough objective measure for the relationships between the different types of freedom - whether it’s political, religious, economical or personal freedom - and other social and economic circumstances. The Human Freedom Index is an annually co-published report by the Cato Institute, the Fraser Institute, and the Liberales Institut at the Friedrich Naumann Foundation for Freedom.

In this lab, you’ll be analyzing data from Human Freedom Index reports from 2008-2016. Your aim will be to summarize a few of the relationships within the data both graphically and numerically in order to find which variables can help tell a story about freedom.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let’s load the packages.

```
library(tidyverse)
library(openintro)
data('hfi', package='openintro')
```

The data

The data we’re working with is in the openintro package and it’s called **hfi**, short for Human Freedom Index.

1. What are the dimensions of the dataset?

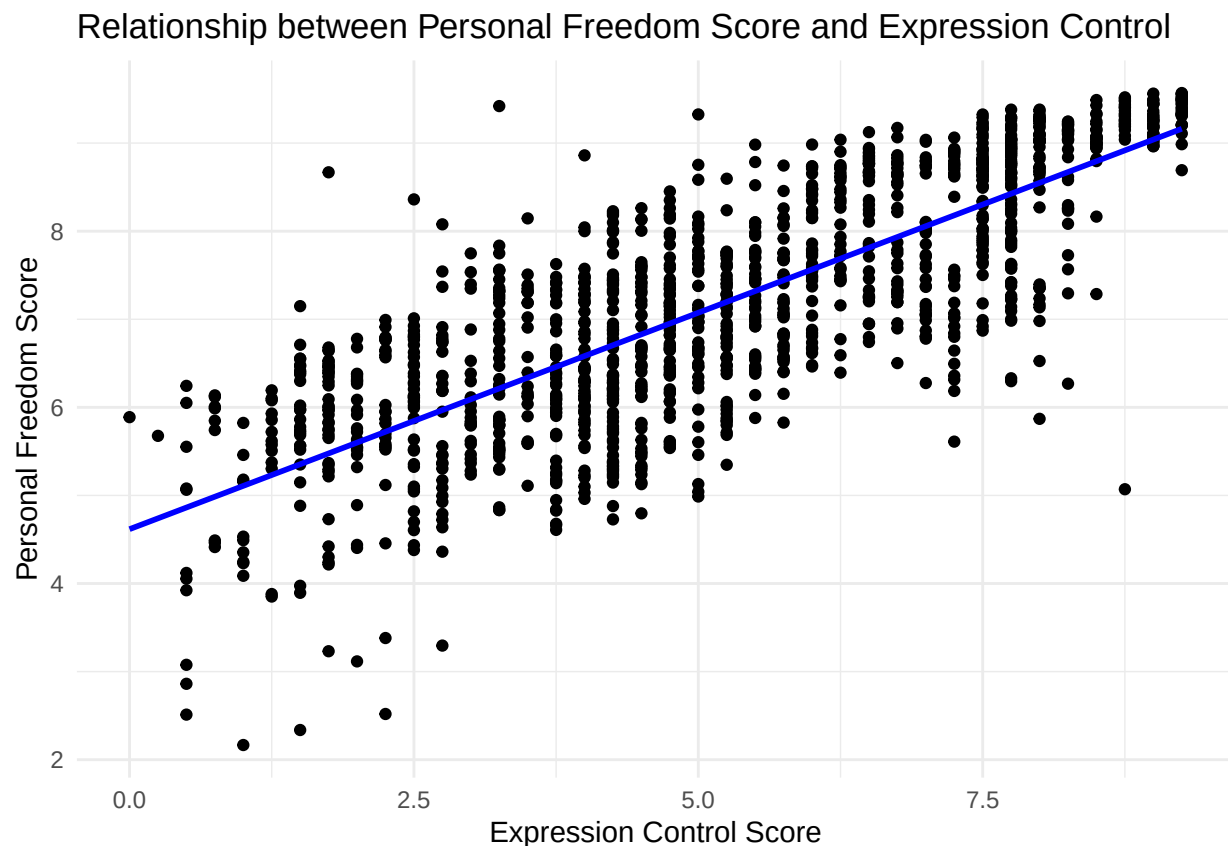
```
?hfi
```

The data set has 1458 observations and 123 variables.

2. What type of plot would you use to display the relationship between the personal freedom score, **pf_score**, and one of the other numerical variables? Plot this relationship using the variable **pf_expression_control** as the predictor. Does the relationship look linear? If you knew a country’s **pf_expression_control**, or its score out of 10, with 0 being the most, of political pressures and controls on media content, would you be comfortable using a linear model to predict the personal freedom score?

To display the relationship between the personal freedom score (pf_score**) and another numerical variable like **pf_expression_control**, we can use a scatterplot. It helps us to visualize the relationship between two numerical variables and is suitable for determining if a linear relationship exists.**

```
# Scatterplot of pf_score vs pf_expression_control
ggplot(hfi, aes(x = pf_expression_control, y = pf_score)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(title = "Relationship between Personal Freedom Score and Expression Control",
       x = "Expression Control Score",
       y = "Personal Freedom Score") +
  theme_minimal()
```



The scatterplot shows us a positive relationship between the Expression Control Score and the Personal Freedom Score. The data points follow a linear trend.

A linear model can be used for the prediction purpose, but it should be interpreted with caution, because we should consider the variability in the data. It would be beneficial to assess the residuals to check if the linear model is an adequate fit or if a more complex model such as another non-linear approach might be needed to better capture the relationship.

If the relationship looks linear, we can quantify the strength of the relationship with the correlation coefficient.

```
hfi %>%
  summarise(cor(pf_expression_control, pf_score, use = "complete.obs"))
```

```
## # A tibble: 1 x 1
##   'cor(pf_expression_control, pf_score, use = "complete.obs")'
##                                                                 <dbl>
## 1                                                                 0.796
```

Here, we set the `use` argument to “complete.obs” since there are some observations of NA.

Sum of squared residuals

In this section, you will use an interactive function to investigate what we mean by “sum of squared residuals”. You will need to run this function in your console, not in your markdown document. Running the function also requires that the `hfi` dataset is loaded in your environment.

Think back to the way that we described the distribution of a single variable. Recall that we discussed characteristics such as center, spread, and shape. It’s also useful to be able to describe the relationship of two numerical variables, such as `pf_expression_control` and `pf_score` above.

3. Looking at your plot from the previous exercise, describe the relationship between these two variables. Make sure to discuss the form, direction, and strength of the relationship as well as any unusual observations.

Form: The relationship appears to be linear, as the data points tend to follow a straight-line pattern. This indicates that a linear model could be suitable for capturing the relationship between `pf_expression_control` and `pf_score`.

Direction: The scatterplot shows a positive relationship as its moving from left to right in the upward direction, it means that as `pf_expression_control` increases, the `pf_score` also tends to increase. This suggests that countries with higher `pf_expression_control` scores tend to have higher personal freedom scores.

Strength: While we can see a clear linear trend, along with that we can also see some variability in the data, which means the strength of the linear relationship is moderate rather than very strong. Some data points are spread away from the line, indicating that the model might not perfectly predict `pf_score` based solely on `pf_expression_control`. But we observed from the above code that the correlation between the two variables is 79%.

Unusual Observations: There are a few data points that lie further away from the main cluster, which could be considered outliers. These observations may suggest that certain countries do not follow the general trend as closely as others, indicating unique situations or factors affecting their freedom scores.

Overall we can say that, the relationship is linear and positive, with a moderate strength, and the presence of some variability and potential outliers should be taken into account when interpreting predictions.

Just as you’ve used the mean and standard deviation to summarize a single variable, you can summarize the relationship between these two variables by finding the line that best follows their association. Use the following interactive function to select the line that you think does the best job of going through the cloud of points.

```
# This will only work interactively (i.e. will not show in the knitted document)
hfi <- hfi %>% filter(complete.cases(pf_expression_control, pf_score))
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score)
```

After running this command, you’ll be prompted to click two points on the plot to define a line. Once you’ve done that, the line you specified will be shown in black and the residuals in blue. Note that there are 30 residuals, one for each of the 30 observations. Recall that the residuals are the difference between the observed values and the values predicted by the line:

$$e_i = y_i - \hat{y}_i$$

The most common way to do linear regression is to select the line that minimizes the sum of squared residuals. To visualize the squared residuals, you can rerun the plot command and add the argument `showSquares = TRUE`.

```
DATA606::plot_ss(x = hfi$pf_expression_control, y = hfi$pf_score, showSquares = TRUE)
```

Note that the output from the `plot_ss` function provides you with the slope and intercept of your line as well as the sum of squares.

4. Using `plot_ss`, choose a line that does a good job of minimizing the sum of squares. Run the function several times. What was the smallest sum of squares that you got? How does it compare to your neighbors?

The smallest sum of squares i got was 1449.45, as i am doing this lab at home i compared it with my other experiments and i saw that the sum of the squares range from 1400's to 16,000's

The linear model

It is rather cumbersome to try to get the correct least squares line, i.e. the line that minimizes the sum of squared residuals, through trial and error. Instead, you can use the `lm` function in R to fit the linear model (a.k.a. regression line).

```
m1 <- lm(pf_score ~ pf_expression_control, data = hfi)
```

The first argument in the function `lm` is a formula that takes the form `y ~ x`. Here it can be read that we want to make a linear model of `pf_score` as a function of `pf_expression_control`. The second argument specifies that R should look in the `hfi` data frame to find the two variables.

The output of `lm` is an object that contains all of the information we need about the linear model that was just fit. We can access this information using the summary function.

```
summary(m1)
```

```
##
## Call:
## lm(formula = pf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8467 -0.5704  0.1452  0.6066  3.2060
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.61707    0.05745   80.36  <2e-16 ***
## pf_expression_control 0.49143    0.01006   48.85  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8318 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.6342, Adjusted R-squared:  0.634
## F-statistic: 2386 on 1 and 1376 DF, p-value: < 2.2e-16
```

Let's consider this output piece by piece. First, the formula used to describe the model is shown at the top. After the formula you find the five-number summary of the residuals. The "Coefficients" table shown next is key; its first column displays the linear model's y-intercept and the coefficient of `pf_expression_control`. With this table, we can write down the least squares regression line for the linear model:

$$\hat{y} = 4.61707 + 0.49143 \times pf_expression_control$$

One last piece of information we will discuss from the summary output is the Multiple R-squared, or more simply, R^2 . The R^2 value represents the proportion of variability in the response variable that is explained by the explanatory variable. For this model, 63.42% of the variability in runs is explained by at-bats.

5. Fit a new model that uses `pf_expression_control` to predict `hf_score`, or the total human freedom score. Using the estimates from the R output, write the equation of the regression line. What does the slope tell us in the context of the relationship between human freedom and the amount of political pressure on media content?

```
# linear model
m2 <- lm(hf_score ~ pf_expression_control, data = hfi)

# summary of the model
summary(m2)

##
## Call:
## lm(formula = hf_score ~ pf_expression_control, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.6198 -0.4908  0.1031  0.4703  2.2933
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.153687   0.046070  111.87  <2e-16 ***
## pf_expression_control 0.349862   0.008067   43.37  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.667 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.5775, Adjusted R-squared:  0.5772
## F-statistic: 1881 on 1 and 1376 DF, p-value: < 2.2e-16
```

Based on the output of the linear model, the equation of the regression line can be written as:

$$hf_score = 5.1537 + 0.3499 \times pf_expression_control$$

Interpretation of the Slope

The slope of the regression line is approximately 0.3499. This indicates that for every one-unit increase in the `pf_expression_control` score (representing more political pressure and controls on media content), the total human freedom score (`hf_score`) is expected to increase by 0.3499 units, on average.

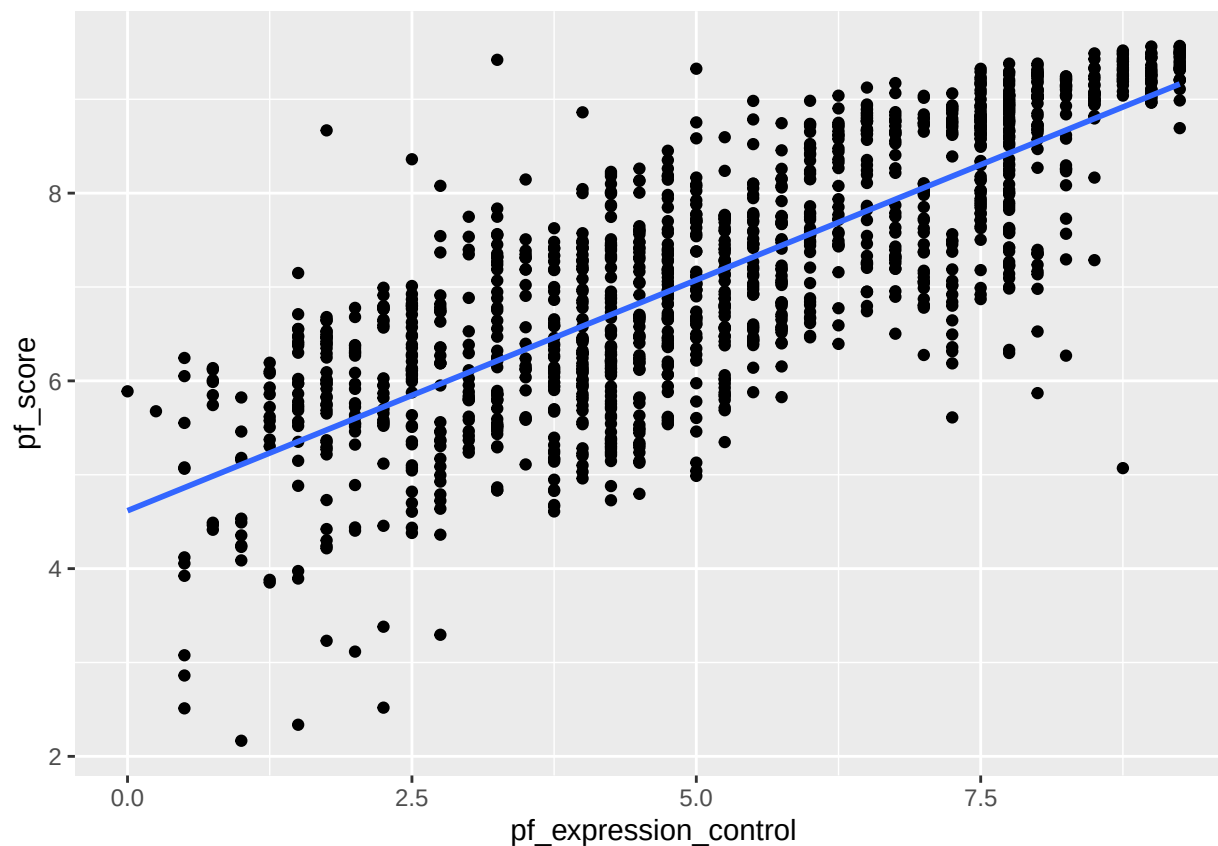
The positive slope suggests that higher scores of `pf_expression_control` (which indicate less political pressure on media content) are associated with higher human freedom scores. Thus, as political pressure and control on media content decrease, the overall human freedom score tends to increase, indicating a positive relationship between freedom of expression and general human freedom.

The R^2 value is 0.5775, that means 57.75% of the variability in the total human freedom score is explained by the political pressure on media content (`pf_expression_control`). This suggests a moderate fit of the linear model to the data.

Prediction and prediction errors

Let's create a scatterplot with the least squares line for `m1` laid on top.

```
ggplot(data = hfi, aes(x = pf_expression_control, y = pf_score)) +  
  geom_point() +  
  stat_smooth(method = "lm", se = FALSE)
```



Here, we are literally adding a layer on top of our plot. `geom_smooth` creates the line by fitting a linear model. It can also show us the standard error `se` associated with our line, but we'll suppress that for now.

This line can be used to predict y at any value of x . When predictions are made for values of x that are beyond the range of the observed data, it is referred to as *extrapolation* and is not usually recommended. However, predictions made within the range of the data are more reliable. They're also used to compute the residuals.

6. If someone saw the least squares regression line and not the actual data, how would they predict a country's personal freedom score for one with a 6.7 rating for `pf_expression_control`? Is this an

overestimate or an underestimate, and by how much? In other words, what is the residual for this prediction?

To predict the personal freedom score (`pf_score`) for a country with a `pf_expression_control` rating of 6.7 using the least squares regression line equation:

we use the regression equation

The equation for the regression line from the previous output is:

$$pf_score = 4.61707 + 0.49143 \times pf_expression_control$$

$$pf_score = 4.61707 + 0.49143 \times 6.7$$

Calculating this:

$$pf_score = 4.61707 + 3.29258 = 7.90965$$

So, the predicted personal freedom score for a country with a `pf_expression_control` rating of 6.7 is approximately 7.91.

Determining if it's an overestimate or an underestimate

To determine if this prediction is an overestimate or an underestimate, we need the actual personal freedom score for a country with a `pf_expression_control` of 6.7 for this we made the below calculations:

```
# Filtering the dataset to find the closest `pf_expression_control` value to 6.7
closest_match <- hfi %>%
  filter(abs(pf_expression_control - 6.7) < 0.1) %>%
  select(pf_expression_control, pf_score)

closest_match
```

```
## # A tibble: 34 x 2
##   pf_expression_control pf_score
##   <dbl>               <dbl>
## 1         6.75         7.43
## 2         6.75         8.22
## 3         6.75         8.77
## 4         6.75         7.87
## 5         6.75         7.39
## 6         6.75         7.25
## 7         6.75         7.79
## 8         6.75         8.27
## 9         6.75         7.19
## 10        6.75         7.75
## # i 24 more rows
```

```
# average pf_score for the closest matches
average_pf_score <- closest_match %>%
  summarise(mean_pf_score = mean(pf_score))
```

```
average_pf_score
```

```
## # A tibble: 1 x 1
##   mean_pf_score
##           <dbl>
## 1           8.01
```

Calculating the residual

The residual is calculated as:

$$\text{Residual} = \text{Actual } pf_score - pf_score^{\wedge}$$

$$\text{Residual} = 8.01 - 7.90 = 0.11$$

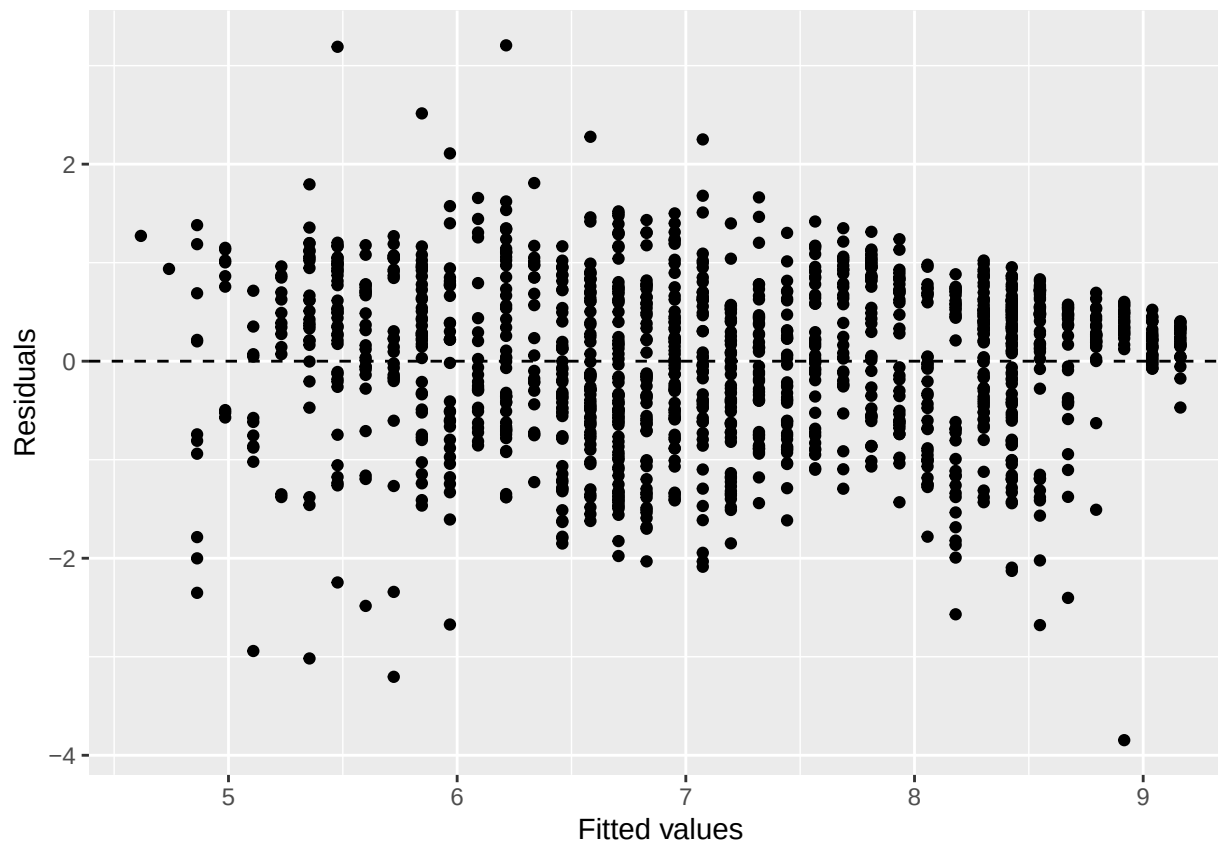
Since the residual is positive (0.11), it indicates that the predicted value is an underestimate of the actual personal freedom score by approximately 0.11 units.

Model diagnostics

To assess whether the linear model is reliable, we need to check for (1) linearity, (2) nearly normal residuals, and (3) constant variability.

Linearity: You already checked if the relationship between `pf_score` and 'pf_expression_control' is linear using a scatterplot. We should also verify this condition with a plot of the residuals vs. fitted (predicted) values.

```
ggplot(data = m1, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  xlab("Fitted values") +
  ylab("Residuals")
```

Notice here that `m1` can also serve as a data set because stored within it are the fitted values ($\hat{y}$) and the residuals. Also note that we're getting fancy with the code here. After creating the scatterplot on the first layer (first line of code), we overlay a horizontal dashed line at $y = 0$ (to help us check whether residuals are distributed around 0), and we also rename the axis labels to be more informative.

7. Is there any apparent pattern in the residuals plot? What does this indicate about the linearity of the relationship between the two variables?

******In the plot we can see that, the residuals are plotted against the fitted values (predicted `pf_score`). Here's what we can observe about the plot:

Analysis:

1. No Apparent Pattern in Residuals:

- The residuals seem to be scattered randomly around the horizontal line at zero, without any clear pattern (e.g., curves, clusters, or systematic structures).
- This randomness indicates that the residuals do not display any particular structure, which suggests that the linearity assumption holds.

2. Linearity Condition:

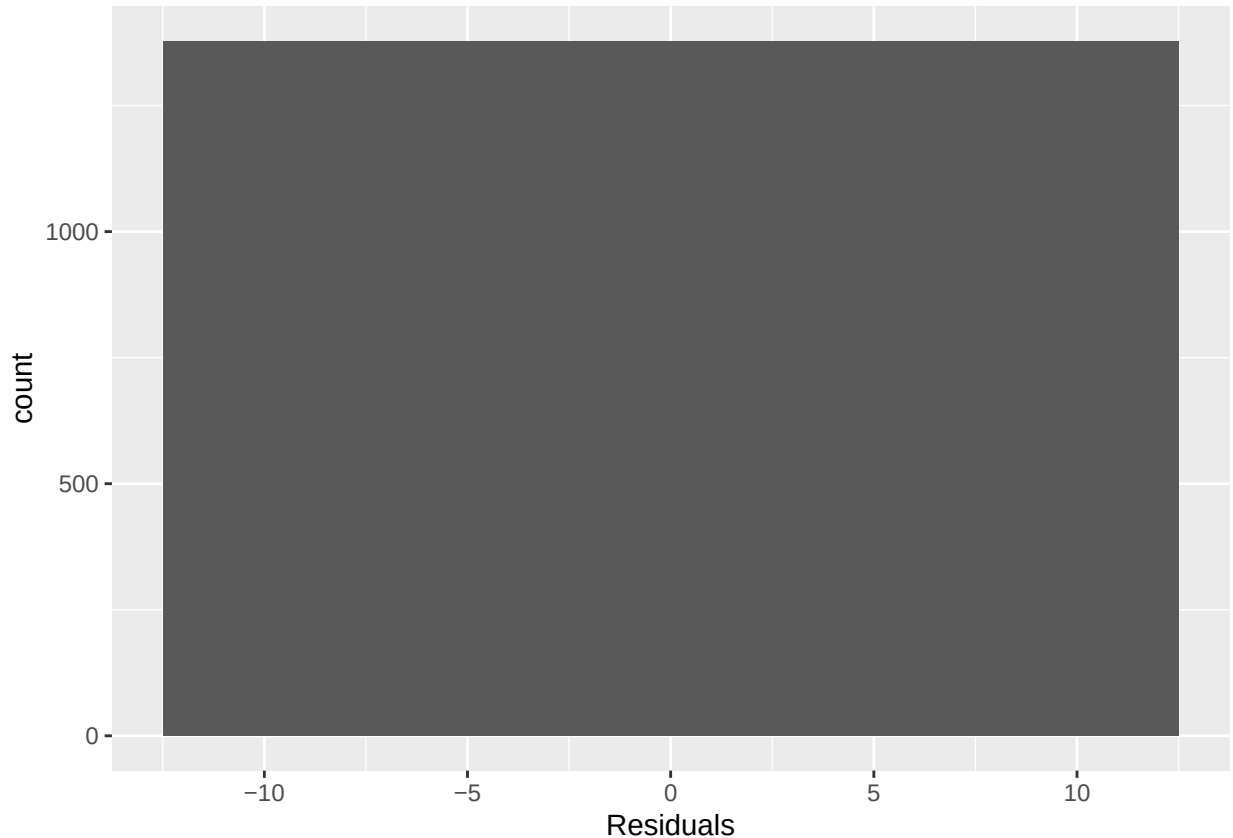
- Since there is no apparent pattern, it suggests that the relationship between `pf_expression_control` and `pf_score` is approximately linear.

Conclusion:

The residual plot indicates that the linear model fits the data well. This suggests that a linear model is appropriate for predicting `pf_score` using `pf_expression_control`.**

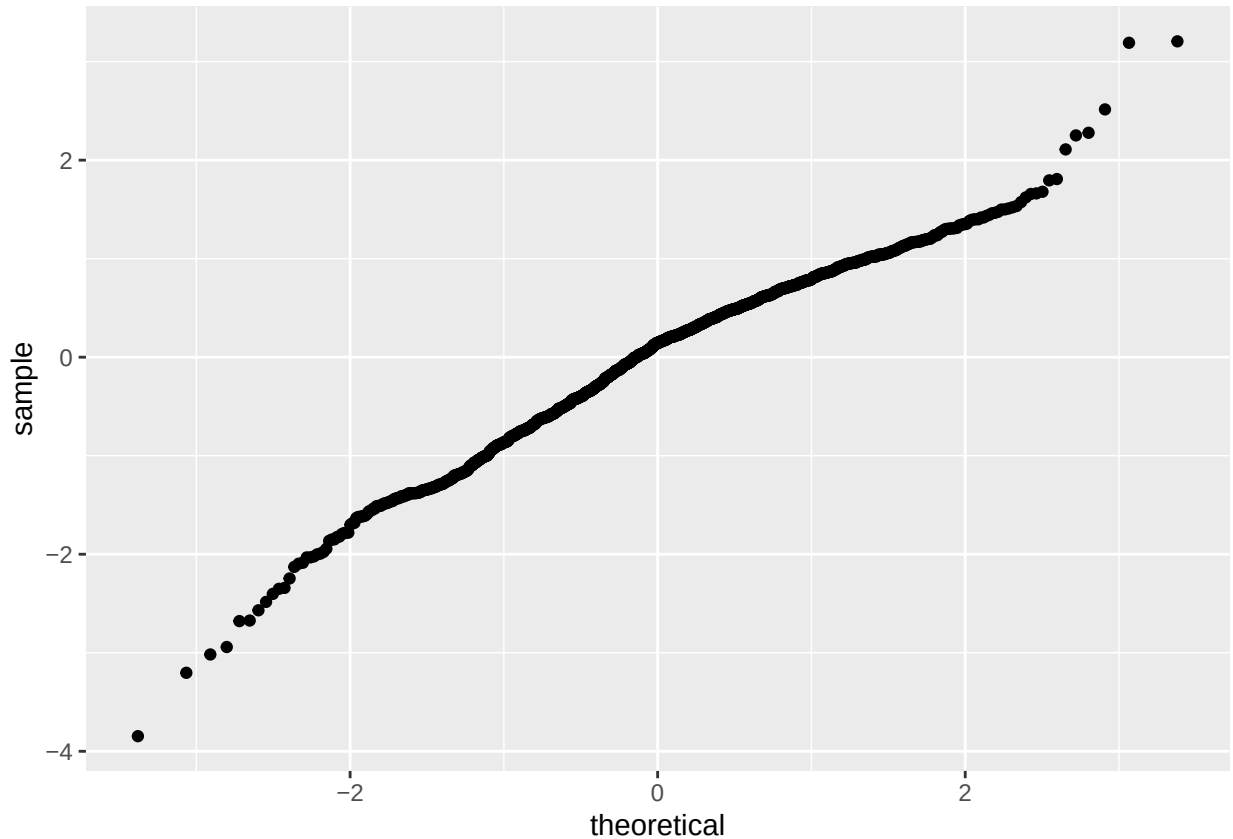
Nearly normal residuals: To check this condition, we can look at a histogram

```
ggplot(data = m1, aes(x = .resid)) +  
  geom_histogram(binwidth = 25) +  
  xlab("Residuals")
```



or a normal probability plot of the residuals.

```
ggplot(data = m1, aes(sample = .resid)) +  
  stat_qq()
```



Note that the syntax for making a normal probability plot is a bit different than what you're used to seeing: we set `sample` equal to the residuals instead of `x`, and we set a statistical method `qq`, which stands for “quantile-quantile”, another name commonly used for normal probability plots.

8. Based on the histogram and the normal probability plot, does the nearly normal residuals condition appear to be met?

Based on the histogram and the normal probability plot (Q-Q plot), we can make the following analysis:

Histogram Analysis

1. The histogram shows a symmetric shape. However, the bars are densely packed, making it difficult to clearly assess if the distribution is perfectly normal.

Q-Q Plot Analysis

1. The Q-Q plot shows that most of the residuals lie close to the straight line, which indicates that they follow a normal distribution reasonably well.
2. There are some deviations at the tails, particularly at the extreme ends, suggesting that there may be some mild departure from normality (potentially a few outliers).

Conclusion

Overall, the nearly normal residuals condition appears to be reasonably met, as the residuals exhibit a roughly symmetric shape in the histogram and follow the line closely in the Q-Q plot. There may be some

mild deviations from perfect normality at the extremes, but these do not seem severe enough to significantly violate the assumption.

Constant variability:

9. Based on the residuals vs. fitted plot, does the constant variability condition appear to be met?

Analysis of the Residuals vs. Fitted Plot:

Spread of Residuals: In the residuals vs. fitted, the residuals appear to be scattered evenly around the horizontal line at zero, with no obvious patterns or funnel-shaped distributions.

Consistent Variability: The spread of the residuals does not appear to systematically increase or decrease across different levels of fitted values. This suggests that the variability is roughly constant.

Conclusion Based on the residuals vs. fitted plot, the constant variability condition seems to be met. The residuals show a consistent spread across the range of fitted values, indicating that the assumption is satisfied for the linear model.

**

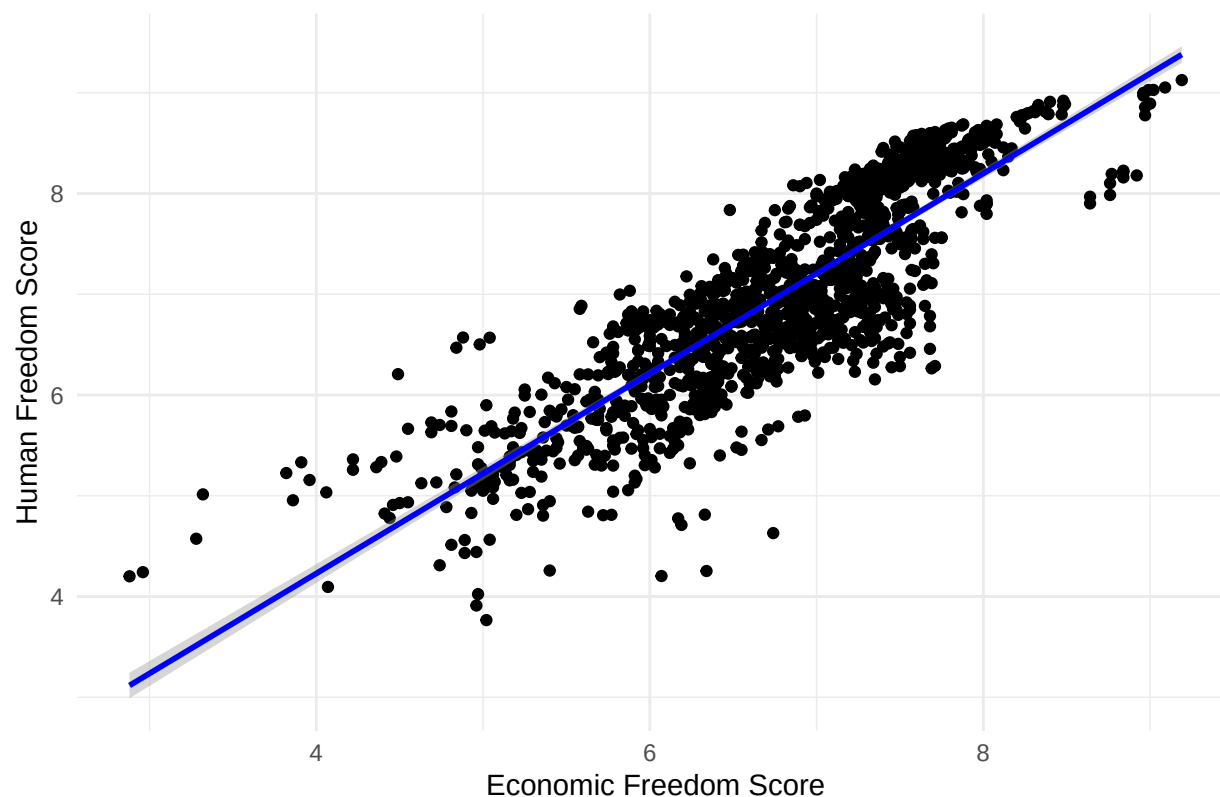
More Practice

- Choose another freedom variable and a variable you think would strongly correlate with it.. Produce a scatterplot of the two variables and fit a linear model. At a glance, does there seem to be a linear relationship?

Here I am choosing the variables `hf_score` (Human Freedom Score) and `ef_score` (Economic Freedom Score), as these two are likely to be correlated since economic freedom is often associated with overall human freedom.

```
# Scatterplot of Human Freedom Score vs. Economic Freedom Score with a linear regression line
ggplot(hfi, aes(x = ef_score, y = hf_score)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  labs(title = "Scatterplot of Human Freedom Score vs. Economic Freedom Score",
       x = "Economic Freedom Score",
       y = "Human Freedom Score") +
  theme_minimal()
```

Scatterplot of Human Freedom Score vs. Economic Freedom Score



```
# Fitting a linear model for Human Freedom Score as a function of Economic Freedom Score
model <- lm(hf_score ~ ef_score, data = hfi)
```

```
# Summary of the model
summary(model)
```

```
##
## Call:
## lm(formula = hf_score ~ ef_score, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.31864 -0.36668  0.05449  0.41767  1.49198
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.25906    0.11112   2.331  0.0199 *
## ef_score     0.99245    0.01624  61.117 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5324 on 1376 degrees of freedom
## (80 observations deleted due to missingness)
## Multiple R-squared:  0.7308, Adjusted R-squared:  0.7306
## F-statistic: 3735 on 1 and 1376 DF, p-value: < 2.2e-16
```

Based on the scatterplot and the linear model output, we can analyze the relationship between `hf_score` (Human Freedom Score) and `ef_score` (Economic Freedom Score):

Scatterplot Analysis:

- The scatterplot shows a strong positive relationship between **Economic Freedom Score** and **Human Freedom Score**. As the `ef_score` increases, the `hf_score` also tends to increase, indicating a linear relationship.
- The data points are relatively close to the fitted line, suggesting a strong correlation.

Linear Model Output:

- **Regression Equation:** The equation for the regression line is:

$$\widehat{hf_score} = 0.25906 + 0.99245 \times ef_score$$

This means that for every one-unit increase in the **Economic Freedom Score**, the **Human Freedom Score** is expected to increase by approximately 0.99245 units, on average.

- **Strength of the Relationship:** The R^2 value is 0.7308, indicating that about 73.08% of the variability in **Human Freedom Score** can be explained by the **Economic Freedom Score**. This is a relatively high R^2 , suggesting a strong linear relationship between the two variables.
- **Significance:** The p-value for the slope ($<2e-16$) is extremely small, indicating that the relationship between `ef_score` and `hf_score` is statistically significant.

Conclusion:

There appears to be a strong linear relationship between the **Economic Freedom Score** and the **Human Freedom Score**.

- How does this relationship compare to the relationship between `pf_expression_control` and `pf_score`? Use the R^2 values from the two model summaries to compare. Does your independent variable seem to predict your dependent one better? Why or why not?

1. Model for `pf_expression_control` vs. `pf_score`:

- The R^2 value was approximately **0.5775**, indicating that about **57.75%** of the variability in the **Personal Freedom Score** (`pf_score`) can be explained by the **Political Expression Control** (`pf_expression_control`).

2. Model for `ef_score` vs. `hf_score`:

- The R^2 value was approximately **0.7308**, indicating that about **73.08%** of the variability in the **Human Freedom Score** (`hf_score`) can be explained by the **Economic Freedom Score** (`ef_score`).

Comparison of the Relationships:

- **Higher R^2 for `ef_score` vs. `hf_score`:** The R^2 value for the relationship between **Economic Freedom Score** and **Human Freedom Score** is higher (0.7308) compared to the R^2 value for the relationship between **Political Expression Control** and **Personal Freedom Score** (0.5775). This indicates that `ef_score` is a better predictor of `hf_score` than `pf_expression_control` is of `pf_score`.

- **Explanation:**

- The higher R^2 value suggests that a larger proportion of the variation in Human Freedom Score can be explained by variations in the Economic Freedom Score. This implies a stronger linear relationship between `ef_score` and `hf_score`, meaning economic freedom may have a more direct impact on overall human freedom than political expression control does on personal freedom.

- What's one freedom relationship you were most surprised about and why? Display the model diagnostics for the regression model analyzing this relationship.

One freedom relationship that is most surprising for me is between freedom of religion (`pf_religion`) and personal freedom (`pf_score`). While it's known that countries with higher levels of religious freedom may also enjoy higher personal freedom, the strength of this relationship may not be as strong as anticipated due to the influence of other factors on personal freedom.

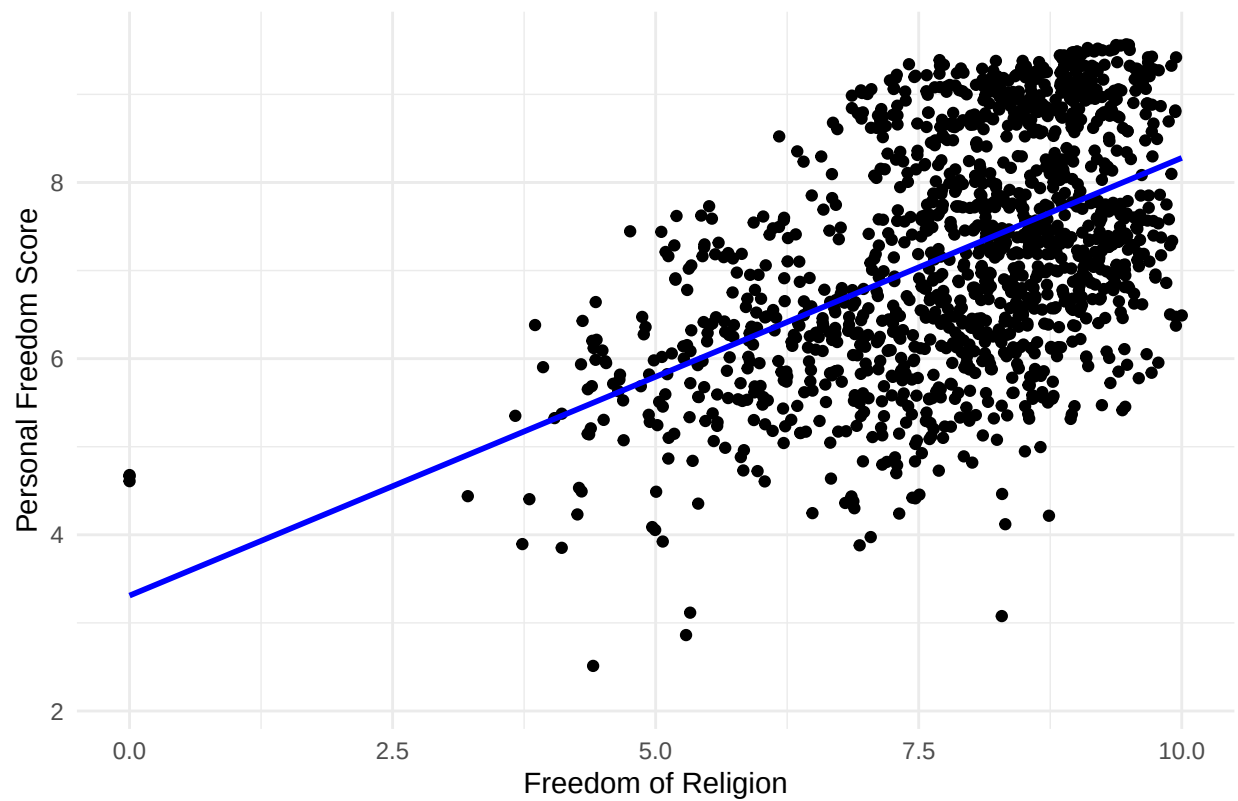
```
# linear model: Freedom of religion vs. Personal freedom score
religion_model <- lm(pf_score ~ pf_religion, data = hfi)
```

```
# Summary of the model
model_summary <- summary(religion_model)
print(model_summary)
```

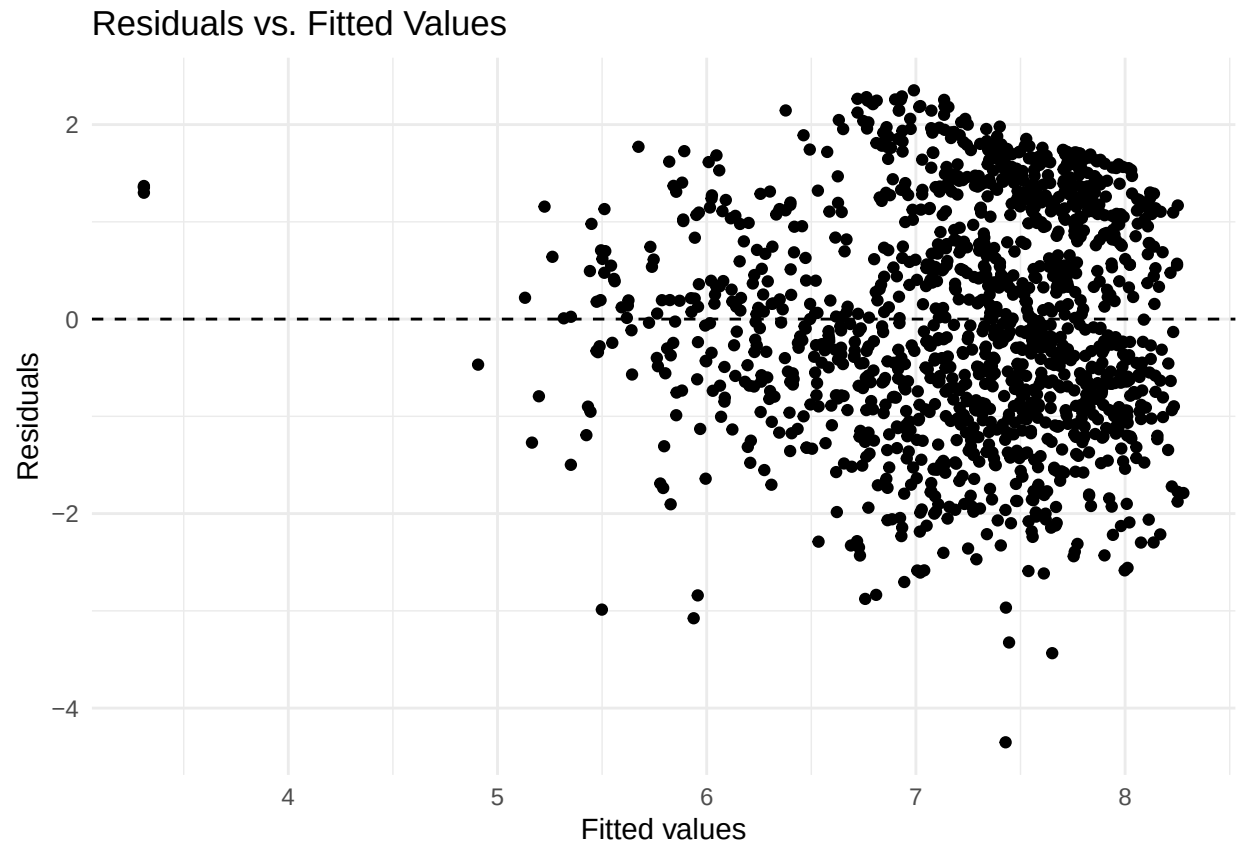
```
##
## Call:
## lm(formula = pf_score ~ pf_religion, data = hfi)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.3515 -0.8728 -0.0744  1.0532  2.3520
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.30981    0.18744   17.66  <2e-16 ***
## pf_religion    0.49689    0.02346   21.18  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.17 on 1366 degrees of freedom
## (90 observations deleted due to missingness)
## Multiple R-squared:  0.2472, Adjusted R-squared:  0.2467
## F-statistic: 448.6 on 1 and 1366 DF, p-value: < 2.2e-16
```

```
# Scatterplot with regression line
ggplot(hfi, aes(x = pf_religion, y = pf_score)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(title = "Scatterplot of Personal Freedom Score vs. Freedom of Religion",
       x = "Freedom of Religion",
       y = "Personal Freedom Score") +
  theme_minimal()
```

Scatterplot of Personal Freedom Score vs. Freedom of Religion

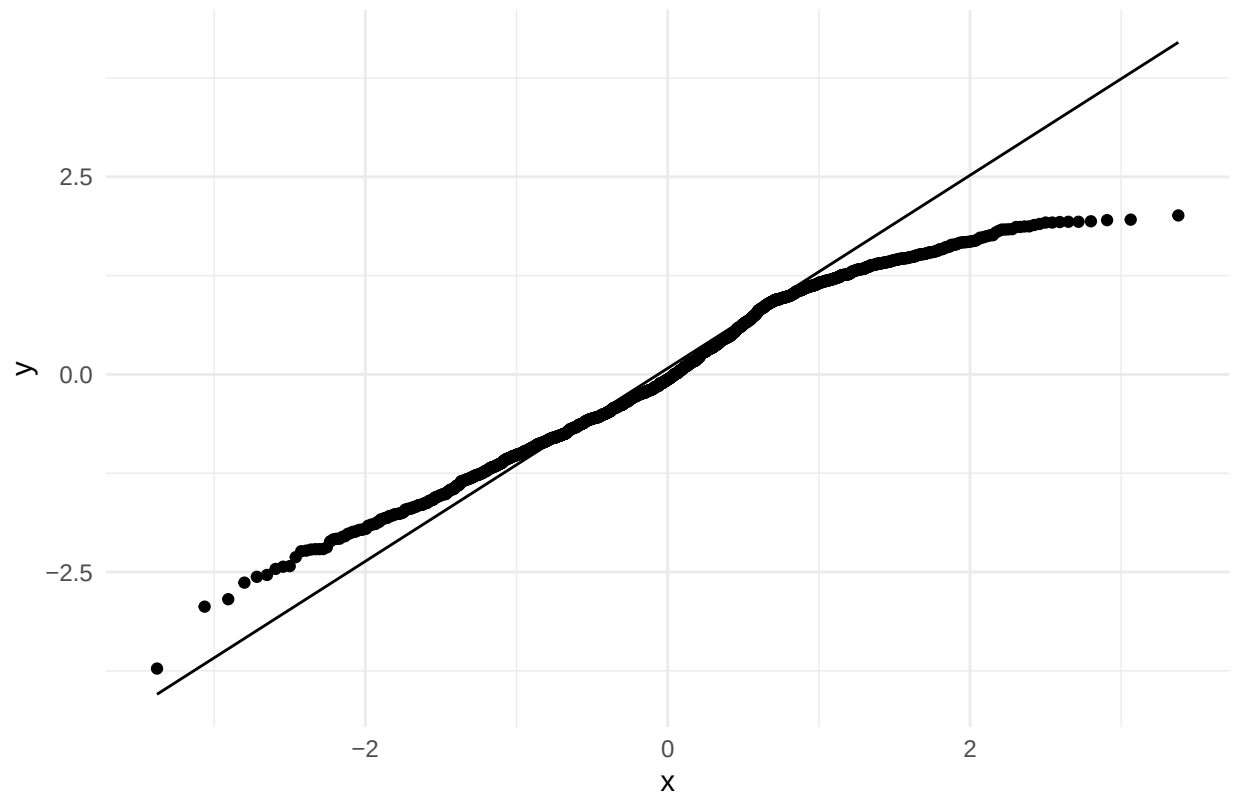


```
# Residuals vs. Fitted plot
ggplot(data = religion_model, aes(x = .fitted, y = .resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Residuals vs. Fitted Values",
       x = "Fitted values",
       y = "Residuals") +
  theme_minimal()
```

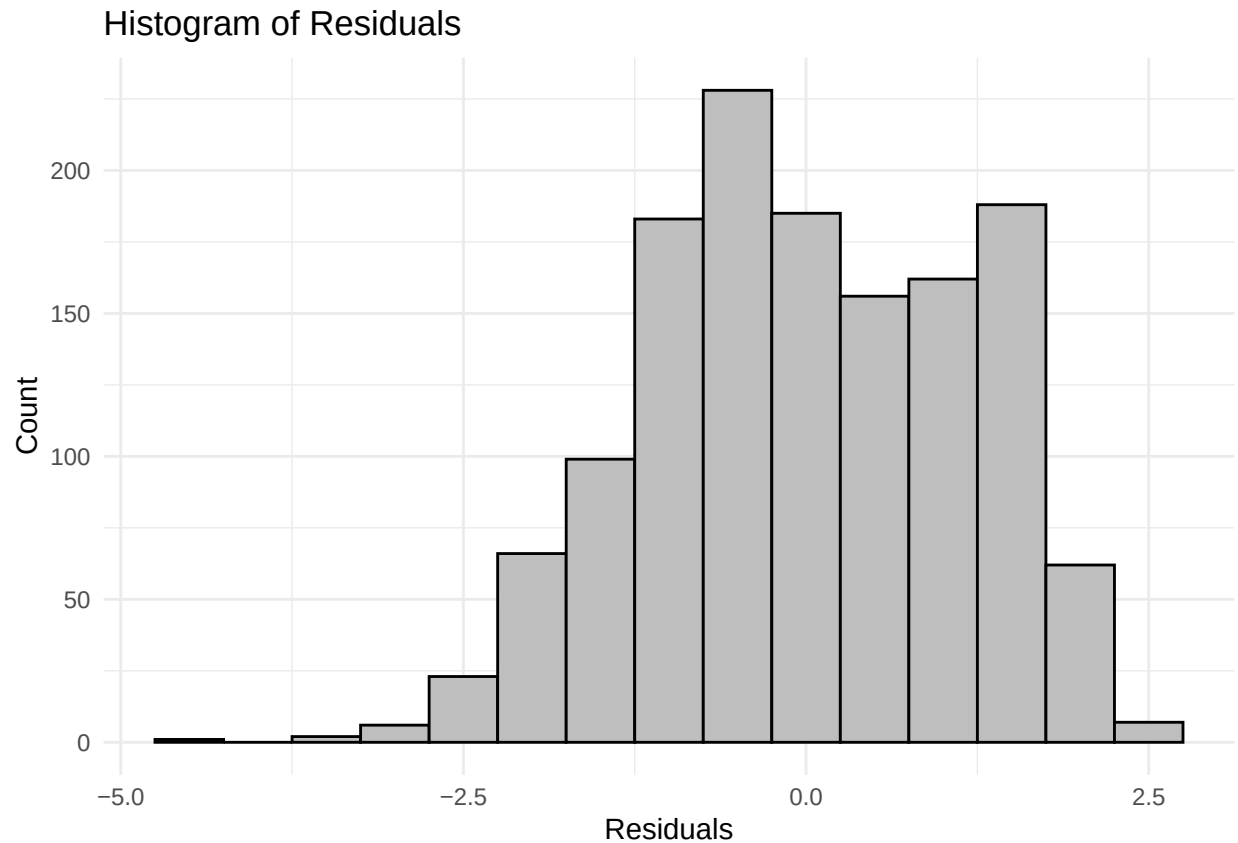



```
# Q-Q plot for normality of residuals
ggplot(data = religion_model, aes(sample = .stdresid)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "Q-Q Plot of Residuals") +
  theme_minimal()
```

Q-Q Plot of Residuals



```
# Histogram of residuals
ggplot(data = religion_model, aes(x = .resid)) +
  geom_histogram(binwidth = 0.5, fill = "grey", color = "black") +
  labs(title = "Histogram of Residuals",
       x = "Residuals",
       y = "Count") +
  theme_minimal()
```



The R^2 value for the linear model relating `pf_score` (Personal Freedom Score) to `pf_religion` (Religious Freedom) is 0.2472. This indicates that approximately 24.72% of the variability in the personal freedom score can be explained by the religious freedom score.

The lower R^2 value here suggests that religious freedom is a less strong predictor of personal freedom compared to economic freedom's relationship with human freedom.

The results indicate that while there is a positive relationship between religious freedom and personal freedom, other factors may also play a significant role in determining the personal freedom score.
